

Project Report

MLB Spending vs Performance

1. Introduction

Major League Baseball teams spend vastly different amounts of money on player salaries, which raises an important question about whether spending more actually leads to better performance. Large-market organizations can spend heavily on payroll, while others operate with much smaller budgets and rely more on player development, scouting, or low-cost roster construction. Because of this difference, payroll is an important factor to study when evaluating team success.

In this project, I analyze the relationship between MLB payroll and team performance during the 2025 season. More specifically, I use payroll data from Spotrac and team performance data from Baseball Reference to determine if higher spending is associated with more wins. I also examine which teams used their money most efficiently and whether the structure of payroll spending, measured through active cash compared to total cash, appears to help or hurt team performance.

The research questions for this project are:

Is there a correlation between payroll and team success?

Which teams spent their money most effectively?

When comparing active cash to total cash, which teams were the most frugal, and did this help or hurt their performance?

2. Data

This project uses two primary sources of data: Spotrac payroll data and Baseball Reference team performance data. Spotrac was used to collect team payroll information for the 2025 MLB season, including active payroll and total cash. Baseball Reference was used to collect team performance statistics, including wins, losses, team ranking, and runs allowed.

The Spotrac dataset included each team's abbreviation, active payroll, and total cash. The payroll columns originally contained dollar signs and commas, so these values were cleaned and converted into numeric variables. The team column also required cleaning

because the team abbreviations appeared twice, such as “NYM NYM.” To fix this, I kept only the first abbreviation for each team.

The Baseball Reference dataset included full team names rather than abbreviations. Since the payroll dataset used abbreviations, I created a dictionary to convert each full team name into its corresponding abbreviation. For example, “New York Yankees” was converted to “NYY” and “Chicago Cubs” was converted to “CHC.” After this cleaning step, the two datasets were merged horizontally using the team abbreviation as the key. The final merged dataset contained 30 observations.

I also created several new variables for analysis. Cost per win was calculated by dividing active payroll by wins. Cash difference was calculated by subtracting active payroll from total cash. Active cash ratio was calculated by dividing active payroll by total cash. These new variables allowed me to compare how much teams spent and how efficiently they used their payroll.

Table 1 Data Dictionary

Column	Type	Source	Description
Team	<i>Text</i>	<i>Spotrac</i>	<i>Abbreviated team name</i>
Team Name	<i>Text</i>	<i>Baseball Reference</i>	<i>MLB team name</i>
Rank	<i>Numeric/Ordinal</i>	<i>Baseball Reference</i>	<i>End of season ranking</i>
Wins	<i>Numeric</i>	<i>Baseball Reference</i>	<i>Total win count</i>
Losses	<i>Numeric</i>	<i>Baseball Reference</i>	<i>Total loss count</i>
Runs Allowed	<i>Numeric</i>	<i>Baseball Reference</i>	<i>Number of runs allowed throughout the season</i>
Active Cash	<i>Numeric</i>	<i>Spotrac</i>	<i>Team payroll tied to active players</i>
Total Cash	<i>Numeric</i>	<i>Spotrac</i>	<i>Team’s total cash payroll obligations</i>
Win Percentage	<i>Numeric</i>	<i>Created Variable</i>	<i>Wins divided by total games</i>
Cost per Win	<i>Numeric</i>	<i>Created Variable</i>	<i>Active payroll divided by wins</i>
Cash Difference	<i>Numeric</i>	<i>Created Variable</i>	<i>Total cash minus active payroll</i>
Active Cash Ratio	<i>Numeric</i>	<i>Created Variable</i>	<i>Active payroll divided by total cash</i>

3. Analysis

3.1 Payroll and Team Success

I first wanted to determine whether teams with higher payrolls tend to win more games. My initial hypothesis was that there would be a positive relationship between payroll and wins because teams with more money should be able to acquire more talented players. To test this, I calculated the correlation between active payroll and wins, as well as the correlation between total cash and wins.

The correlation between active payroll and wins was 0.58, which indicates a moderate positive relationship. The correlation between total cash and wins was 0.48, which is also positive but slightly weaker. This suggests that teams with higher payrolls tended to win more games, but the relationship was not strong enough to conclude that payroll alone determines success.

I also created a scatterplot comparing active payroll and wins (figure 1). The graph shows a general upward trend which means teams with higher active payrolls often had more wins. However, there are also clear exceptions. For example, some teams spent large amounts of money but did not finish near the top of the league, while some lower-payroll teams were still competitive. This supports the idea that payroll matters, but it is not the only factor that explains team success.

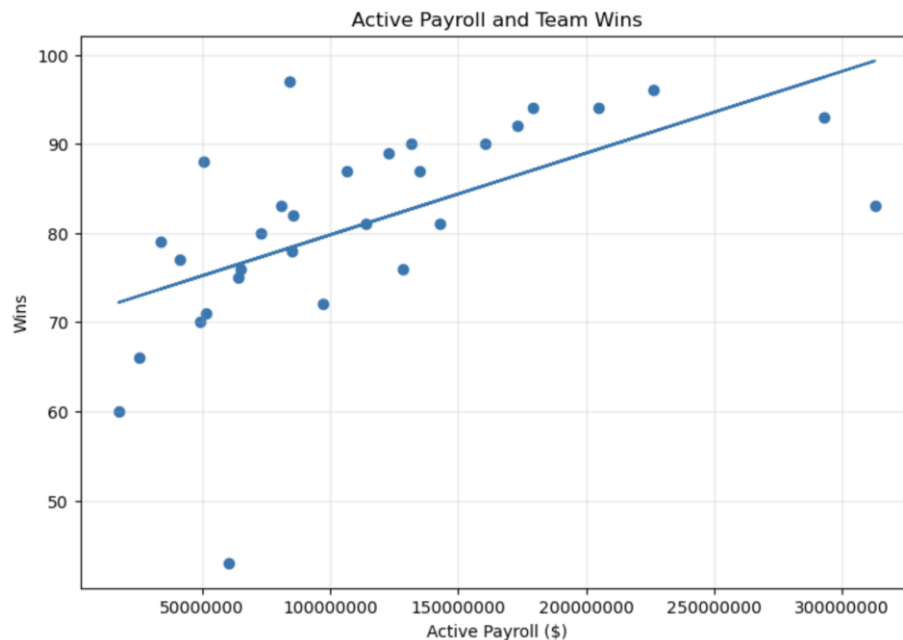


Figure 1: Active Payroll vs Team Wins

3.2 Spending Efficiency

Next, I wanted to determine which teams spent their money most effectively. My hypothesis was that the most efficient teams would not necessarily be the teams with the highest payrolls. Instead, I expected some teams with lower payrolls to produce strong win totals relative to the amount of money they spent. To analyze spending efficiency, I created a new variable called cost per win (figure 2). This was calculated by dividing each team's active payroll by its number of wins. A lower cost per win means that a team spent less money on each win, making it more efficient.

The teams with the lowest cost per win included the Chicago White Sox, Washington Nationals, Miami Marlins, Tampa Bay Rays, and Cleveland Guardians. The White Sox had the lowest cost per win mainly because their active payroll was very low, not because they were one of the most successful teams. This shows the key limitation of the cost per win metric: it will reward teams for spending very little, even if they do not perform well.

Because of this limitation, I also grouped teams into payroll tiers (figure 3). The low-payroll teams averaged about 70.5 wins, the middle-payroll teams averaged about 82.5 wins, and the high-payroll teams averaged about 90.0 wins. This gives us a clearer picture of the relationship between spending and performance. While some low-payroll teams were efficient, the teams with higher payrolls performed better on average.

Overall, the results show that efficiency is not just about spending the least. A team can have a low cost per win and still be unsuccessful if it wins very few games. The best examples of effective spending are teams that combine reasonable payroll levels with strong performance, such as teams that achieve above-average wins without being one of the highest spenders.

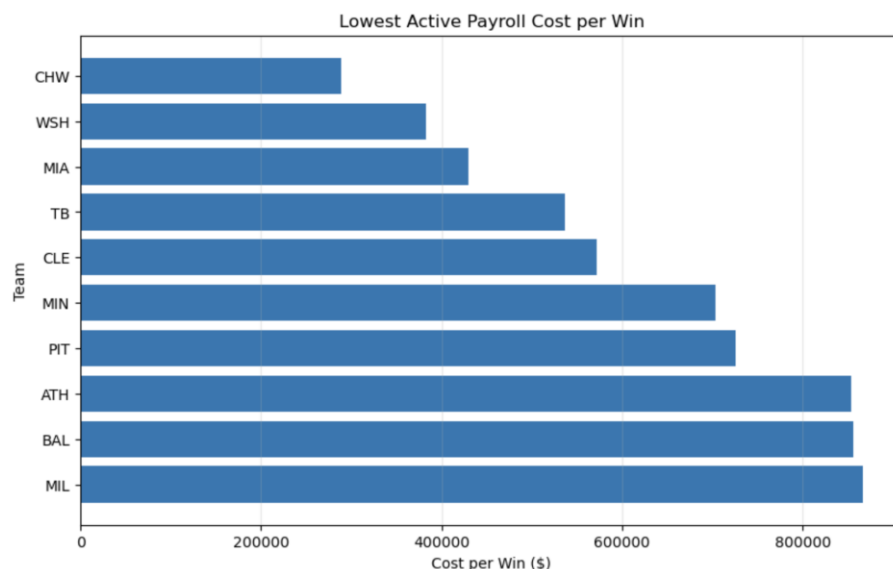


Figure 2: Lowest Active Payroll per Win

	Payroll Tier	Average_Wins	Average_Active_Payroll	Number_of_Teams
0	Low Payroll	70.5	45841248.2	10
1	Middle Payroll	82.5	97819534.1	10
2	High Payroll	90.0	195948571.3	10

Figure 3: Wins Compared to Payroll Tier

3.3 Active Cash, Total Cash, and Frugality

Finally, I examined the difference between active cash and total cash. Active cash represents money spent on players currently contributing to the active roster, while total cash includes total payroll obligations (many teams are still paying contracts on players no longer on their team). My hypothesis was that teams with a higher active cash ratio would perform better because a larger share of their payroll was going toward active players.

To analyze this question, I calculated active cash ratio by dividing active payroll by total cash. A lower ratio means that a smaller share of a team's total cash was tied to active payroll, while a higher ratio means that more of the team's cash spending was connected to active roster players.

The teams with the lowest active cash ratios included the Washington Nationals, Chicago White Sox, Baltimore Orioles, Minnesota Twins, and Arizona Diamondbacks. These teams had a relatively small percentage of their total cash going toward active payroll. In some cases, this appeared to be connected with weaker performance. For example, the Nationals and White Sox both had low active cash ratios and finished with low win totals.

The correlation between active cash ratio and wins was approximately 0.671, which indicates a moderately strong positive relationship. I also grouped teams into low, middle, and high active cash share groups (Figures 4,5). The low active share group averaged about 71 wins, the middle group averaged about 81 wins, and the high active share group averaged about 91 wins. This suggests that teams with more of their total cash directed toward active payroll tended to perform better.

This does not prove that active cash ratio directly causes winning, but it does suggest that payroll structure matters. Teams that had more of their cash tied to active players generally performed better than teams with a lower active payroll share. This may indicate that teams benefit when a larger portion of their financial resources is being used on players who are currently contributing to wins.

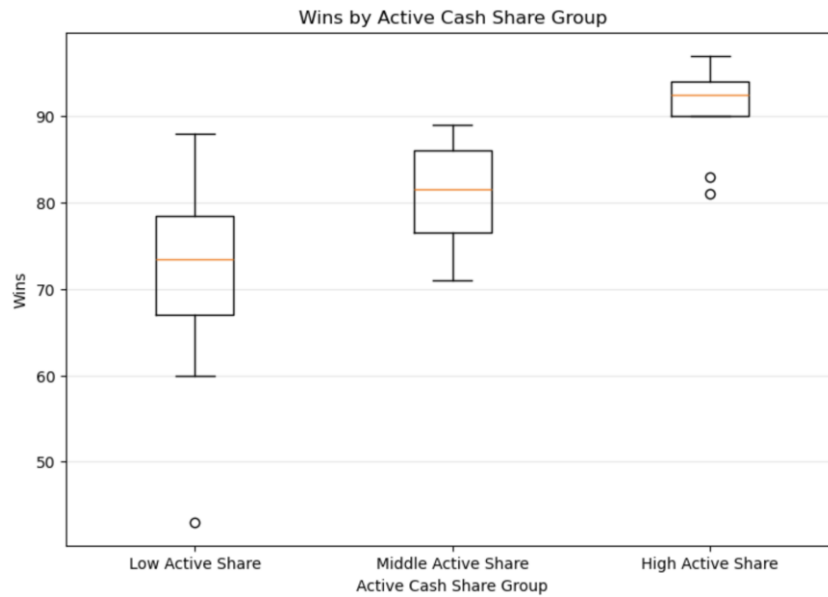


Figure 4: Wins by Active Cash Share

	Active Cash Group	Average_Wins	Average_Active_Cash_Ratio	Number_of_Teams
0	Low Active Share	71.0	0.338055	10
1	Middle Active Share	81.0	0.543299	10
2	High Active Share	91.0	0.692828	10

Figure 5: Average Cash Group Summary

4. Conclusion

In this project, I analyzed the relationship between MLB payroll and team success during the 2025 season. Overall, the results show that money matters, but it does not fully determine success. Active payroll had a positive relationship with wins, meaning higher-spending teams generally performed better. However, spending efficiency also needed context. Some low-payroll teams had a low cost per win, but that did not always mean they were successful. For example, the White Sox appeared efficient by cost per win but only won 60 games. The payroll tier analysis gave a clearer picture, as high-payroll teams averaged more wins than middle- and low-payroll teams.

The strongest result came from active cash ratio. Teams with a higher share of total cash going toward active payroll tended to win more games. This suggests that not only the amount of money spent matters, but also how that money is allocated.

This project is limited because it only uses one season of data and does not include injuries, trades, coaching, or player performance. Future work could analyze multiple seasons or include more detailed player statistics to better understand what makes payroll spending effective.